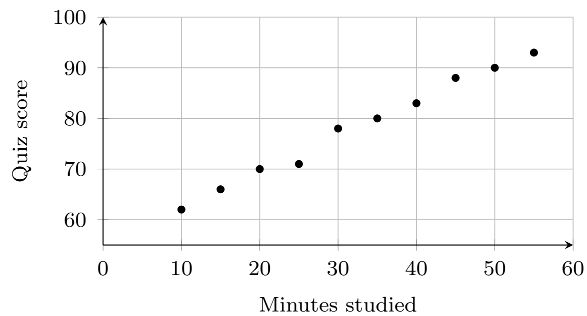
Scatter Plots and Informal Lines of Fit

Reading a scatter plot, describing the association, drawing a line of fit and writing its equation, and predicting from it

Directions

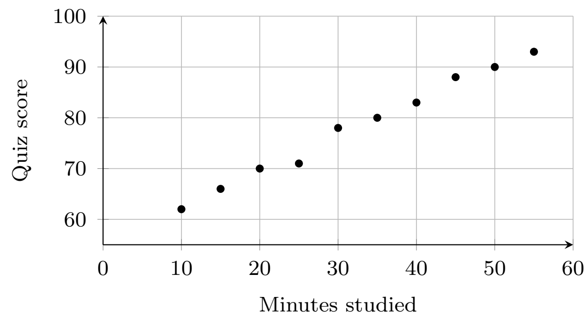
- A **line of fit** is a straight line drawn by eye through the middle of the cloud of points — not a line joining two dots, and not a line through every point.
- When you predict from a line, write the substitution before the answer.
- Give the units in your written answers, and round to one decimal place where a value does not come out exact.

1. The scatter plot shows how long each of ten students revised and the quiz score they earned. How many points did the student who revised for 30 minutes score?



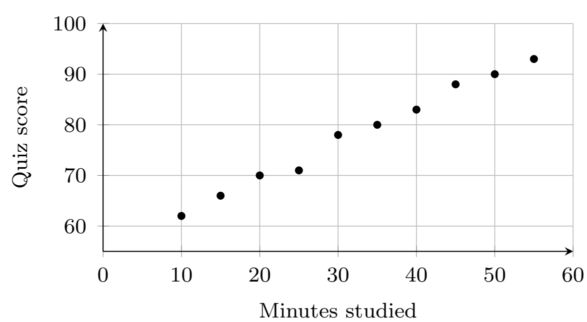
Answer: ________________

2. Using the same data, describe the association between revision time and quiz score. Say whether it is positive or negative, whether it is strong or weak, and whether it looks linear. Then write the highest quiz score in the plot.



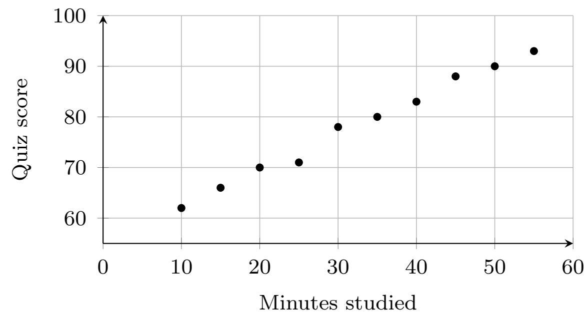
Answer: ________________

3. Draw a line of fit on the plot below: a straight line that passes as close as possible to as many points as possible, with roughly as many points above it as below it. Draw yours so that it passes through $(10, 62)$ and $(50, 90)$, then find the slope of that line.



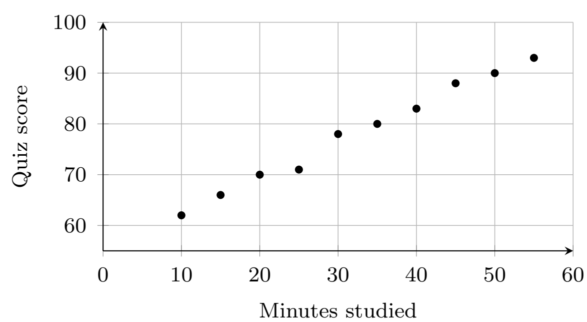
Answer: ________________

4. Your line of fit passes through $(10, 62)$ and has slope 0.7. Substitute that point into $y = mx + b$ to find b, then write the full equation of the line of fit.



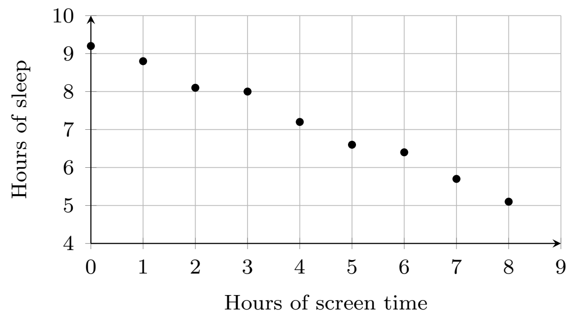
Answer: ________________

5. Use your line of fit $y = 0.7x + 55$ to predict the quiz score of a student who revises for 60 minutes. Show the substitution, and say whether this is an interpolation or an extrapolation.



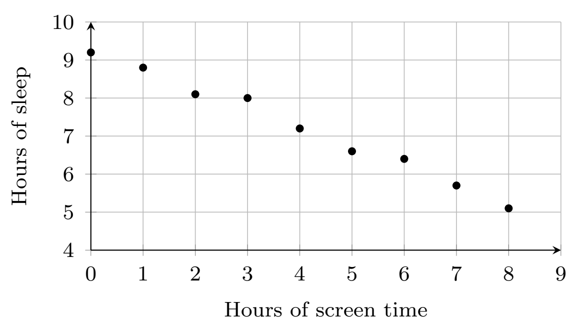
Answer: ________________

6. This scatter plot shows the screen time and the sleep of nine students on the same night. How many hours of sleep did the student with 5 hours of screen time get?



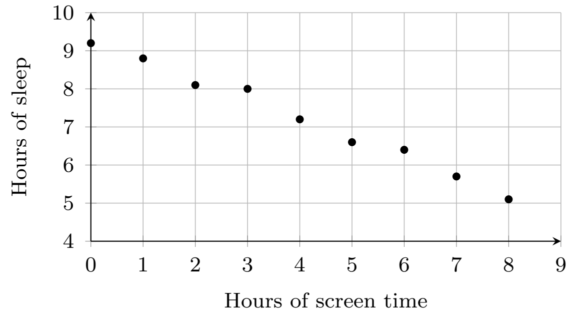
Answer: ____________________

7. Using the same data, describe the association between screen time and sleep in words, and write the smallest number of hours of sleep shown in the plot.



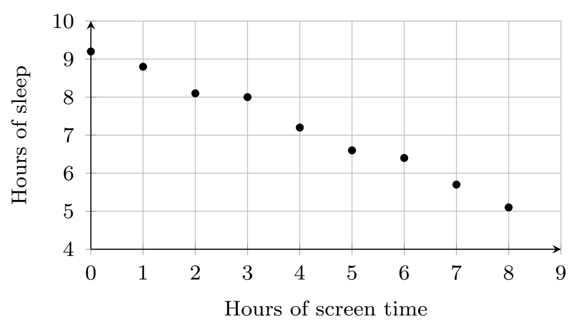
Answer: ____________________

8. A line of fit for this data passes through $(0, 9)$ and $(8, 5)$. Find its slope, write the equation of the line, and then say in a sentence what the slope means for these students.



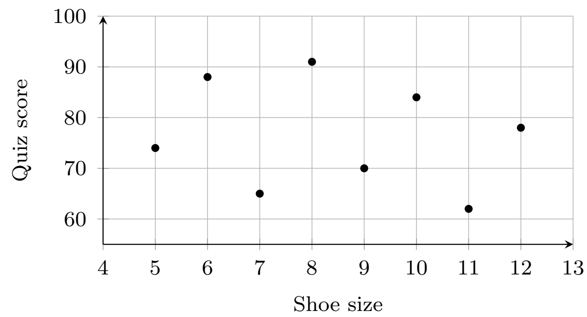
Answer: ____________________

9. Use the line of fit $y = -0.5x + 9$ to predict the sleep of a student with 7 hours of screen time. Then read the actual value for 7 hours off the plot. Which is larger, and by how much?



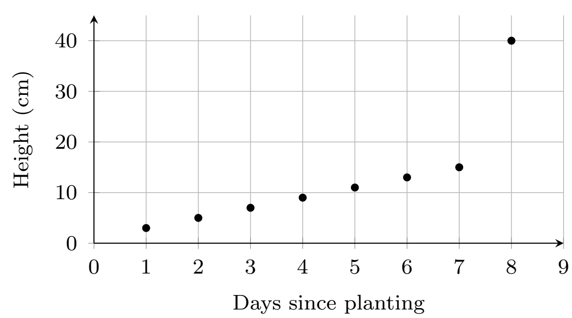
Answer: ____________________

10. This scatter plot pairs each student's shoe size with their quiz score. Would a line of fit be a useful summary here? Explain what the plot shows, then compute the mean quiz score for the eight students.



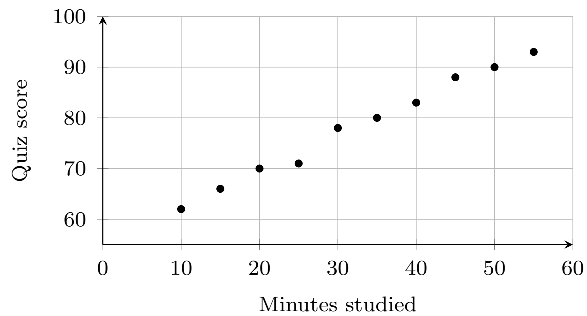
Answer: ________________

11. A gardener measured one plant on eight days. One of the eight points is an outlier. Name the outlier, then compute the range of all eight heights and the range of the seven heights without it. Explain how the outlier would change a line of fit drawn through all eight points.



Answer: ________________

12. Return to the revision data and its line of fit $y = 0.7x + 55$. Use the line to predict the score of a student who revises for 200 minutes. Compute the prediction, then explain why nobody should trust it.



Answer: ___________________